GLOBE SCOTT MOTORS · FREE ENGINEERING RESOURCE

Motor Premature Failure Prevention Checklist

For maintenance teams and specifying engineers responsible for industrial PMDC, BLDC and geared motors

Complete Section 1 when a motor is commissioned, then repeat Sections 2 to 4 at your planned maintenance interval and compare every reading with the baseline. Use Section 5 when specifying a new or replacement motor, and Section 6 whenever a motor fails.

Section 1 — Commissioning baseline

- [ ] Record nameplate data: voltage, current, speed, duty type (S1/S2/S3), insulation class and IP code.
- [ ] Measure insulation resistance at 500 V DC (windings below 1 kV) and correct it to 40 °C.
- [ ] Record running current at a known, repeatable load.
- [ ] Record vibration velocity RMS at each bearing in horizontal, vertical and axial directions.
- [ ] Record frame and bearing housing temperatures together with the ambient at the motor.
- [ ] Record alignment readings and belt tension as installed.

Section 2 — Routine electrical checks

- [ ] Insulation resistance corrected to 40 °C: investigate any sustained downward trend.
- [ ] Polarisation index on larger windings: at least 2.0 for Class B and higher insulation (IEEE 43).
- [ ] Running current at the same load: investigate any rising trend.
- [ ] Terminal voltage under load, measured at the motor rather than at the panel.
- [ ] Re-torque terminals; inspect cable glands, gaskets and seals before the monsoon.
- [ ] Test insulation resistance before restarting any motor that has stood idle or been exposed to water.

Section 3 — Routine mechanical checks

- [ ] Bearing and frame temperature: investigate a rise of 10 K or more over baseline at comparable load and ambient.
- [ ] Vibration: trend against ISO 20816 zones and act on step changes.
- [ ] Relubricate at the specified interval, shorten it for hot bearings and never over-grease.
- [ ] Alignment within about 0.05 mm after any intervention; belt tension to the drive maker's value.
- [ ] Check fixings and soft foot; clear fins, vents and fan covers of dust.
- [ ] Standby motors: keep dry and turn shafts periodically to avoid false brinelling.

Section 4 — Brushed DC motors

- [ ] Measure brush length and replace before the manufacturer's minimum length.
- [ ] Check brush spring pressure and that brushes slide freely in their holders.
- [ ] Inspect the commutator film: uniform chocolate brown is healthy; streaks, grooves or burns are not.
- [ ] Remove carbon dust from the commutator and brush gear.
- [ ] Look for sources of oil mist or silicone vapour near the motor.

Section 5 — Selection and specification

- [ ] Duty type matches the real cycle and the motor is sized on RMS torque with margin.
- [ ] Ambient at the motor and site altitude stated; Class F insulation with a Class B rise.
- [ ] IP code chosen for the actual dust, washdown or water-logging exposure.
- [ ] Bearing L10h at the real radial and axial loads ≥ required service life; overhung load within rating.
- [ ] Supply tolerance, ripple and transients defined; bearing-current mitigation specified for inverter drives.
- [ ] Brushed or brushless technology chosen on running hours and maintenance access.
- [ ] Supplier test scope confirmed: surge, HV dielectric, dynamometer load test, endurance and temperature rise.

Section 6 — Failure investigation (RCFA)

- [ ] Preserve the failed bearing, a grease sample and photographs of windings and commutator before cleaning.
- [ ] Collect operating data: load, current, temperature, vibration and supply history.
- [ ] Identify the mechanism from the evidence using the table below.
- [ ] Identify the condition that accelerated it: installation, environment, supply or selection.
- [ ] Correct the condition, not just the part, and record the action for review.

Failure evidence → mechanism → first check

Evidence	Mechanism	First check
Flaking on bearing raceways	Fatigue, often from overload	Belt tension, overhung load, alignment
Dry, darkened grease	Lubrication failure	Bearing temperature, grease type and interval
Dents, rust, abrasive wear	Contamination or moisture	Seals, IP code, storage conditions
Washboard fluting on raceways	Electrical erosion	Inverter drive, shaft grounding
Darkened, brittle insulation	Thermal ageing	RMS load, ambient, cooling paths
Localised winding burn	Dielectric breakdown	IR history, moisture, surges
Grooved or burnt commutator	Commutation distress	Load, brush grade, contamination
Shaft crack at keyway	Bending fatigue	Overhung load, alignment, resonance

Thresholds are typical engineering guidance. Always apply the motor manufacturer's limits and the applicable standards (IEEE 43, ISO 281, ISO 20816) for the specific machine.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.